

Markscheme

Mathematics: analysis and approaches
Practice question bank

108. (a)

In the expansion of $(1 + ax)^5$, the coefficient of x^2 is 90, where $a > 0$.

Find the value of a .

$$\text{coefficient of } x^2: \binom{5}{2}a^2 = 10a^2 \quad (M1)$$

$$10a^2 = 90 \Rightarrow a^2 = 9 \quad \mathbf{A1}$$

$$\text{since } a > 0: \quad (M1)$$

$$a = 3 \quad \mathbf{A1}$$

[5 marks]